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Impact of anti-anaerobic antibiotic activity on graft-versus-host disease in allogeneic hematopoietic stem cell transplant (HSCT) recipients at an institution that utilizes antibiotic cycling

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Abstract

Background: At our institution, antibiotic cycling for febrile neutropenia is utilized to increase heterogeneity of antibiotic exposure in patients who have undergone an allogeneic hematopoietic stem cell transplantation (allo-HSCT). Development of acute graft-versus-host disease (aGVHD) has been associated with low diversity within stool microbiota. To date, discordant outcomes have been reported implicating anti-anaerobic antibiotic use with the development of aGVHD, and there is currently a lack of published data available in an antibiotic cycled environment. The objective of this study was to determine if there is a difference in the rate of aGVHD in patients who receive anti-anaerobic cycled antibiotics compared with other cycled antibiotics. **Methods:** This was a retrospective, observational study evaluating rates of aGVHD in patients who received antibiotics with anaerobic vs non-anaerobic coverage post-allo-HSCT from January 2008 to January 2018. Univariate and multivariable analyses were performed to assess associations with aGVHD. Secondary outcomes include rate of all stages of aGVHD, progression-free survival, overall survival, 100-day treatment-related mortality (TRM), and 1-year TRM.

Results: A total of 273 patients were included in the study. Baseline characteristics were similar between groups, except patients who received anti-anaerobic antibiotics had more unrelated donors ($P = .002$), were more likely to get myeloablative preparatory regimens ($P = .009$), had less subtherapeutic calcineurin inhibitor serum concentrations ($P = .001$), and more often received T-cell depletion ($P = .004$). The incidence of grades II–IV aGVHD post-HSCT in patients who received anti-anaerobic antibiotics was 32.6% compared with 18.8% in patients who received other antibiotics ($P = .015$). Multivariable analysis showed that the occurrence of grades II–IV aGVHD was associated with cytomegalovirus reactivation (OR = 2.1, 95% CI = 1.0–4.5, $P = .047$), unrelated donors (OR = 6.1, 95% CI = 2.3–16.6, $P < .001$), and use of anti-anaerobic antibiotics (OR = 2.3, 95% CI = 1.1–4.8, $P = .021$). A 100-day TRM in patients who received anti-anaerobic antibiotics was 9.6% compared with 3.6% in patients who received other antibiotics ($P = .046$). One-year TRM in patients who received anti-anaerobic antibiotics was 25.2% compared with 13.8% in patients who

received other antibiotics ($P = .017$). There was no statistically significant difference seen between groups in progression free survival or overall survival.

Conclusion: Variability in baseline characteristics limits ability to make strong conclusions, but patients who received antibiotics with anaerobic coverage during the first 30 days after an allogeneic HSCT appeared to be at an increased risk of developing aGVHD and TRM. Larger well-controlled trials are warranted to further clarify these relationships.

KEYWORDS

allogeneic hematopoietic stem cell transplantation, anaerobic, antibiotic use, graft-versus-host disease

1 | INTRODUCTION

Patients who receive an allogeneic hematopoietic stem cell transplant (allo-HSCT) are at risk for graft-versus-host disease (GVHD). They are also at high risk for infection while neutropenic, and often requiring broad-spectrum antibiotics when becoming febrile or infected. At our institution, an empiric antibiotic cycling rotation is utilized for febrile neutropenia to ensure heterogeneity of antibiotic use. Piperacillin/tazobactam, meropenem, and cefepime are rotated every 3 months as the agent of choice. All three of these antibiotics are effective for the management of febrile neutropenia because of their broad-spectrum, including anti-pseudomonal, activity. However, one difference between the agents is their anti-anaerobic activity, which cefepime lacks.

Animal models have shown that intestinal microflora play a key role in the pathophysiology of GVHD.¹⁻³ Based on this concept, some transplant centers utilize gut decontamination to prevent acute graft-versus-host disease (aGVHD), where antibiotics are used to eradicate intestinal bacteria. Although some studies have reported decreased rates of aGVHD following gut decontamination, this practice remains controversial, and evidence is inconclusive.^{1,4-9}

Recent studies have suggested that having a diverse microbiota is critical to maintaining immune homeostasis.^{1,10-13} Shono and colleagues performed a study in humans and mice that reported increased GVHD-related mortality after exposure to broad-spectrum antibiotics following allo-HSCT.¹⁴ The authors retrospectively reviewed 857 patients post-transplantation and found that treatment of febrile neutropenia with piperacillin/tazobactam and imipenem/cilastatin was significantly associated with increased GVHD-related mortality. In contrast, this increased risk was not seen in patients treated with cefepime or aztreonam.¹⁴ In addition, after mice were exposed to imipenem/cilastatin, there was a significant loss of mucosal lining of the colon, and the intestinal barrier was compromised. This study demonstrated a potential risk of using antibiotics with anti-anaerobic activity linked to acute colonic GVHD after transplant.¹⁴ Given that an antibiotic rotation is used at our institution and includes antibiotics that do and do not cover anaerobes, we created a study with the objective of assessing and comparing the risk of

aGVHD in allo-HSCT patients who received treatment for febrile neutropenia with specific antibiotics.

2 | METHODS

This retrospective study was approved by the institutional review board of West Virginia University. All patients who received an allo-HSCT between January 2008 and December 2017 were included in this study. Patients were divided into two groups: those who received anaerobic coverage and those who did not. If a patient received at least two consecutive doses of an antibiotic with anti-anaerobic activity within the first 30 days post-transplantation, they were placed in the anaerobic coverage group. All other patients were placed in the non-anaerobic group. Agents defined as having anaerobic coverage included carbapenems (meropenem and ertapenem at our institution), piperacillin/tazobactam, ampicillin/sulbactam, amoxicillin/clavulanate, cefoxitin, clindamycin, and metronidazole.

The primary endpoint of this study was to compare the occurrence of grades II–IV aGVHD in each group at 100 days post-transplantation. All suspected cases of aGVHD were histologically confirmed, at least in one target organ. Consensus Conference Criteria and the National Institutes of the Health Consensus Development Project Criteria were used for the grading of acute and chronic GVHD respectively.^{15,16} Secondary endpoints included rate of all stages of aGVHD, progression-free survival (PFS), overall survival (OS), 100-day treatment-related mortality (TRM), and 1-year TRM. Cause of death was documented according to the Center for International Blood and Marrow Transplant Research cause of death coding criteria. Univariate and multivariable analyses were performed to identify risk factors associated with aGVHD. All risk factors included in the univariate analysis were also included in the multivariable analysis. Cytomegalovirus (CMV) reactivation was defined as any viremia greater than 137 IU/mL as identified by our laboratory.

Biostatistical analysis was performed using SAS, Ver 5.0 (Cary). Chi-squared test was used for categorical variables. Wilcoxon's rank test was used for continuous baseline variables. PFS and OS were

estimated using the Kaplan–Meier method. OS was defined as the time from transplantation to death from any cause, and date of last follow-up was used for surviving patients. PFS was calculated using death of any cause and disease progression as events. A 100-day and 1-year TRM was defined as any expiration of a patient while in remission at 100 and 365 days post-transplantation, respectively.

3 | RESULTS

Two hundred and seventy-three patients were included in this study. The anaerobic group contained 135 patients, and the non-anaerobic group contained 138 patients. Baseline patient and allograft characteristics for each group are presented in Table 1. All patients received levofloxacin prophylaxis from Day 2 until neutrophil engraftment or initiation of antibiotics for febrile neutropenia, whichever occurred first. Excluding levofloxacin prophylaxis, 30 patients (22%) in the anaerobic group also received at least one non-anti-anaerobic antibiotic targeted at gram-negative organisms at some point during the first 30 days post-HSCT. These other antibiotics administered to patients in this cohort included cefepime (16.3%), ciprofloxacin (5.2%), ceftriaxone (2.2%), aztreonam (1.5%), cefuroxime (1.5%), ceftazidime (0.7%), and ceftaroline (0.7%), with some patients receiving more than one listed agent. For additional information regarding antibiotics received by patients in each cohort throughout the study period, see Figure 1.

The incidence of grades II–IV aGVHD was significantly lower in the non-anaerobic group compared with the anaerobic group (18.8% [26/138] vs 32.6% [44/135], $P = .014$, respectively). Grades III–IV aGVHD was also observed significantly less in the non-anaerobic group (6.5% [9/138] vs 15.6% [21/135], $P = .012$). The incidence of all grades (I–IV) aGVHD was significantly lower in the non-anaerobic group compared with the anaerobic group (30.4% [42/138] vs 46.8% [63/135], $P = .008$, respectively). The rate of all grade gastrointestinal aGVHD was significantly lower in the non-anaerobic group (5.8% [8/138] vs 20% [27/135], $P = .0005$). Grades III–IV gastrointestinal aGVHD was also observed significantly less in the non-anaerobic group compared with the anaerobic group (4.4% [6/138] vs 11.9% [16/135], $P = .023$, respectively).

Other risk factors associated with aGVHD were also assessed using univariate and multivariable analyses. The univariate analysis showed that in addition to anaerobic antibacterial use, CMV reactivation and having an unrelated donor were associated with grades II–IV aGVHD. The multivariable analysis (Table 2) showed that CMV reactivation (OR = 2.134, 95% CI = 1.01–4.51, $P = .047$) and having an unrelated donor (OR = 6.135, 95% CI = 2.262–16.636, $P < .001$) were associated with grades II–IV aGVHD. Other patients that appeared to be at risk for grades II–IV aGVHD included those who had a diagnosis of acute myeloid leukemia or lymphoma and those with low-risk disease (according to The American Society for Blood and Marrow Transplantation disease-risk categories). A Cox-regression analysis was completed accounting for unrelated donor type and CMV reactivation, using competing risk data analysis, with anaerobic

coverage still showing to be a significant variable with hazard ratio of 1.78 ($P = .017$; Table 3).

The Kaplan–Meier analysis showed no statistically significant difference in PFS (Figure 2) and OS (Figure 3) between groups. Two- and 3-year PFS in the non-anaerobic group is 47% (95% CI = 40.1–57.1%) and 39% (95% CI = 32–49.2%), respectively, compared with 2- and 3-year PFS in the anaerobic group of 41.3% (95% CI = 33.6–50.8%) and 37.4% (95% CI = 29.7–47%), respectively. Two- and 3-year OS in the non-anaerobic group is 54.2% (95% CI = 46.3–63.3%) and 48.6% (95% CI = 40.6–58.1%), respectively, compared with 2- and 3-year OS in the anaerobic group of 47.1% (95% CI = 39.2–56.7%) and 41.3% (95% CI = 33.4–51.1%), respectively. Death caused by GVHD was observed in 5.8% (8/138) of patients in the non-anaerobic group and 12.6% (17/135) of patients in the anaerobic group ($P = .052$). There was no significant difference in the rate of 1-year disease relapse observed in patients in the non-anaerobic group compared with those in the anaerobic group (29.7% [41/138] vs 20% [27/135], $P = .064$, respectively). A 100-day TRM was observed in 3.6% (5/138) of patients in the non-anaerobic group and 9.6% (13/135) of patients in the anaerobic group ($P = .046$). One-year TRM was observed in 13.8% (19/138) of patients in the non-anaerobic group and 25.2% (34/135) of patients in the anaerobic group ($P = .017$).

4 | DISCUSSION

In this retrospective study, we sought to determine the impact of antibiotic spectrum on the development of aGVHD at Day 100 when antibiotics were used within the first 30 days after an allo-HSCT. This is one of the first studies to look at risk of aGVHD based on antibiotic use, specifically using anti-anaerobic activity as a surrogate for gut dysbiosis. Our study found a significant difference between the two groups, leading us to believe that patients who receive antibiotics with anaerobic coverage may be at increased risk for developing aGVHD. This is consistent with what other recent studies have reported.^{13,17} Our rates of grades II–IV aGVHD at Day 100 in each group (19% and 33%) were within commonly reported ranges.^{12,18}

Although most of the baseline characteristics were similar between groups, there were several significant differences. The anaerobic group contained some risk factors for GVHD with more patients receiving transplantation from unrelated donors (62% vs 41%, $P = .002$) and myeloablative preparatory regimens (57% vs 41%, $P = .009$). However, there were also baseline characteristic imbalances that would provide a protective effect from GVHD, including higher incidence of T-cell depletion (64% vs 46%, $P = .004$) and less occurrences of subtherapeutic calcineurin inhibitor serum concentrations (2 vs 4, $P = .001$) in the anaerobic group.^{12,18}

There still remains a lot of unanswered questions about the duration of therapy of anaerobic coverage that would be required to impact aGVHD rates. Also, it is unclear if there are time periods around transplantation that have more or less effect. In one study, Weber and colleagues found that early exposure to antibiotics from Days 7

TABLE 1 Baseline patient, donor, and allograft characteristics

Patient characteristics	Did not receive anaerobic coverage (N = 138)	Received anaerobic coverage (N = 135)	P value
Median patient age, years (range)	51 (17–72)	54 (19–72)	.117
Male gender, (%)	73 (53)	79 (59)	.416
Median ECOG performance status, (range)	1 (0–4)	1 (0–3)	.051
Median HCT-Cl, (range)	2 (0–8)	2 (0–8)	.517
Diagnosis, (%)			
Acute myeloid leukemia	64 (46)	57 (42)	.281
Acute lymphoblastic leukemia	9 (7)	16 (12)	
Lymphoma	30 (22)	22 (16)	
Myelodysplastic syndrome	21 (15)	19 (14)	
Others	14 (10)	21 (16)	
ASBMT disease-risk categories ^a , (%)			
Low risk	52 (40)	56 (46)	.608
Intermediate risk	41 (31)	35 (29)	
High risk	37 (28)	31 (25)	
Very high risk	1 (1)	0 (0)	
Ablative conditioning, (%)	56 (41)	77 (57)	.009
Non-ablative conditioning, (%)	82 (59)	58 (43)	
Received DLI, (%)	10 (7)	11 (8)	.958
Received T-cell depletion, (%)	63 (46)	86 (64)	.004
CNI/MTX, (%)	85 (62)	80 (59)	.794
CNI/MMF, (%)	52 (38)	53 (39)	
Other, (%)	1 (1)	2 (1)	
Median tacrolimus concentration, ng/mL (range)	7.6 (0–14.15)	7.93 (0–12.5)	.292
Median number of subtherapeutic tacrolimus (<5 ng/mL) or cyclosporine (<100 ng/mL) concentrations, (range)	4 (0–16)	2 (0–15)	.001
Allograft characteristics			
Peripheral blood graft, (%)	109 (79)	111 (82)	.719
Related matched (RM), (%)	59 (43)	32 (24)	
Unrelated matched (UM), (%)	57 (41)	84 (62)	.002
Cords, (%)	20 (14)	19 (14)	
Haploidentical, (%)	2 (1)	0 (0)	
ABO mismatched transplant, (%)	63 (56)	66 (61)	.502
Donor and/or recipient CMV seropositive, (%)	112 (81)	109 (82)	.99
CMV reactivation ^b , (%)	66 (48)	61 (45)	.752

CMV, cytomegalovirus; ECOG, Eastern Cooperative Oncology Group; HCT-Cl, hematopoietic cell transplantation-specific comorbidity index; ASBMT, American Society for Blood and Marrow Transplantation; DLI, donor lymphocyte infusion; MTX, methotrexate; CNI, calcineurin inhibitor; MMF, mycophenolate mofetil.

^aBlood. June 5, 2014; 123(23): 3664–3671.

^bDefined as any CMV DNAemia greater than 137 IU/mL as identified by our laboratory.

to 0 significantly reduced abundance of commensal *Clostridiales* and was significantly associated with higher GVHD-related TRM compared with late antibiotic exposure on Day 0 or thereafter.¹⁷ Our study did not specifically look at the association of early antibiotic exposure prior to transplantation and outcomes. Because of inter-individual variability of gastrointestinal microbiome, there is no clear time to onset or duration of dysbiosis caused by antibiotic use.¹⁹

However, previous studies have described human gut dysbiosis persisting for up to a year after a short course of an antibiotic.¹⁹

Given that aGVHD is associated with several patient and allograft characteristics, we performed a multivariable analysis to determine the independent risk factors associated with this outcome. Anaerobic coverage was a significantly associated independent risk factor for the development of grades II–IV aGVHD. As shown

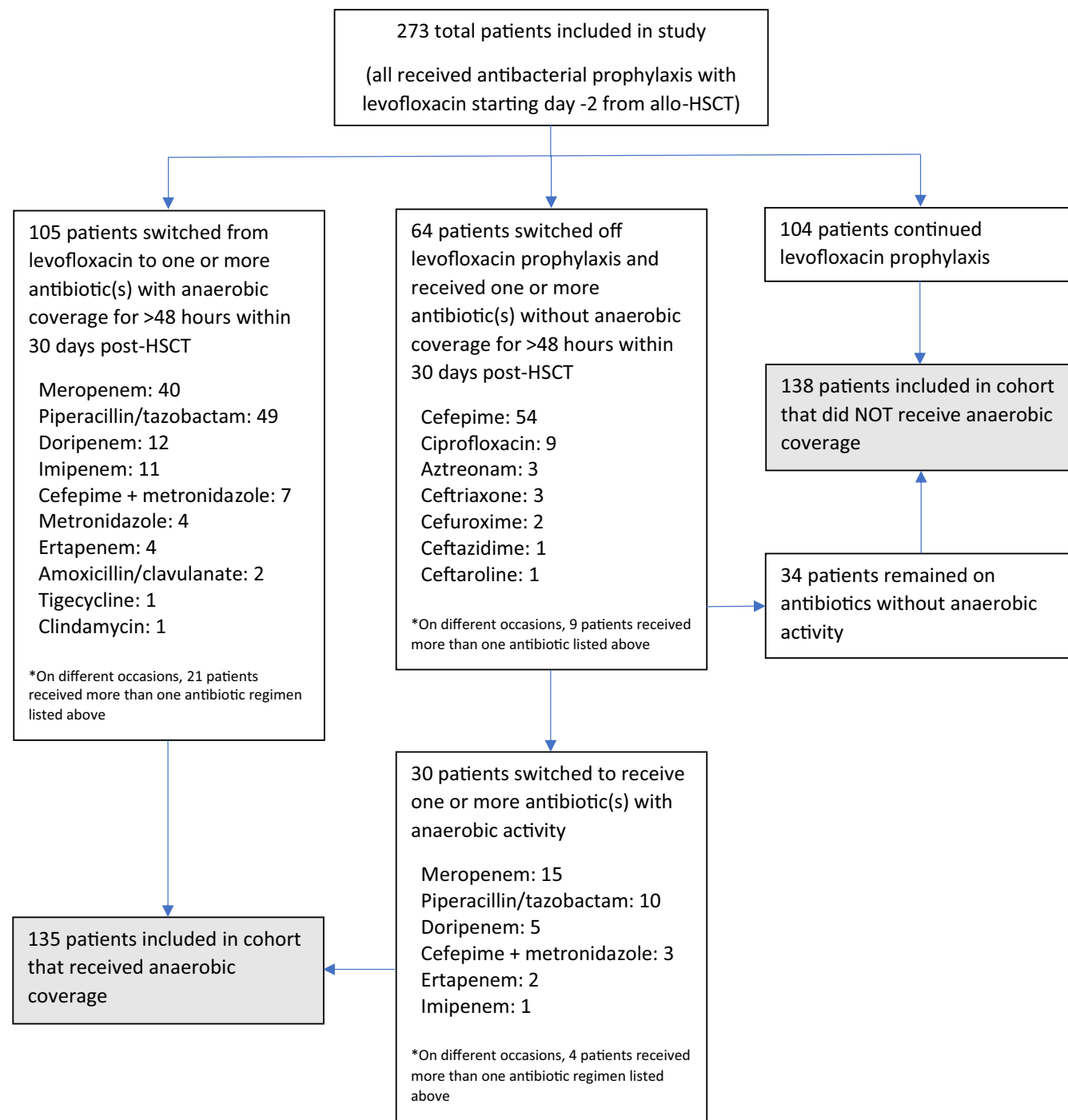


FIGURE 1 Flow diagram of study cohort and antibiotics received throughout study period

in Table 2, patients with CMV reactivation and patients who received an unrelated donor transplant were associated with grades II–IV aGVHD. Both risk factors are known to increase the rate of aGVHD.^{12,18,20}

A recently published study by Nishi and colleagues reported somewhat divergent results at their academic medical center in Japan, concluding that fourth generation cephalosporins had a significant increase on occurrence of grades II–IV aGVHD.²¹ The authors acknowledged that these findings were different than those found in other studies done in the United States. A couple of

explanations for this were provided by the authors, including racial differences affecting the microbiota and differences in antibiotic use. In Japan, cefepime is the first-line antibiotic for neutropenic fever, and used more frequently, while our institution rotates broad-spectrum antibiotics without clinician preference, or bias based on patient condition. In addition, our study specifically compared antibiotics with anaerobic coverage vs non-antianaerobic antibiotics in a head-to-head fashion, while Nishi and colleagues chose to look at the risk of aGVHD associated with each antibiotic independently. This particular population in Japan also did not receive prophylaxis

TABLE 2 Multivariable analysis of characteristics associated with aGVHD grade II+

Patient characteristics	Odds ratio	95% Confidence interval	P value
Anaerobic coverage	2.349	1.139–4.841	.021
Age at transplantation	1.001	0.973–1.031	.926
Male gender	1.023	0.498–2.103	.950
ECOG performance score	0.668	0.394–1.132	.134
HCT-CI	1.042	0.859–1.263	.677
Diagnosis			
Acute lymphoblastic leukemia (reference)	1	—	—
Acute myeloid leukemia	0.231	0.073–0.729	.013
Lymphoma	0.142	0.034–0.587	.007
Myelodysplastic syndrome	0.284	0.064–1.261	.098
Others	0.470	0.106–2.076	.319
ASBMT disease-risk categories ^a			
Low risk	0.366	0.148–0.905	.030
Intermediate risk	0.410	0.160–1.049	.063
High risk and very high risk (reference)	1	—	—
Non-ablative conditioning	0.749	0.337–1.664	.478
Received DLI	1.823	0.529–6.280	.341
Received T-cell depletion	0.449	0.187–1.083	.075
CNI/MMF (reference)	1	—	—
CNI/MTX	0.475	0.186–1.210	.119
Other	10.080	0.054–1892.911	.387
Median tacrolimus concentration	0.948	0.617–1.456	.807
Subtherapeutic tacrolimus or cyclosporine concentrations	0.988	0.887–1.100	.820
Allograft characteristics			
Stem cell source			
Bone marrow (reference)	1	—	—
Cords	2.972	0.331–26.718	.331
Peripheral blood	1.245	0.206–7.533	.811
Unrelated matched (UM)	6.135	2.262–16.636	<.001
Donor and/or recipient CMV seropositive	2.173	0.749–6.303	.153
CMV reactivation ^b	2.134	1.010–4.509	.047

CMV, cytomegalovirus; ECOG, Eastern Cooperative Oncology Group; HCT-CI, hematopoietic cell transplantation-specific comorbidity index; ASBMT, American Society for Blood and Marrow Transplantation; DLI, donor lymphocyte infusion; MTX, methotrexate; CNI, calcineurin inhibitor; MMF, mycophenolate mofetil.

^aBlood. June 5, 2014; 123(23): 3664–3671.

^bDefined as any CMV DNAemia greater than 137 IU/mL as identified by our laboratory.

TABLE 3 Cox-regression analysis accounting for donor type and CMV reactivation, using competing risk data analysis

Variables	Hazard ratio	95% CI lower	95% CI upper	P value
Unrelated donor	3.1	1.81	5.32	<.0001
CMV reactivation	1.68	1.05	2.7	.031
Anaerobic coverage	1.78	1.11	2.86	.017

CI, confidence interval; CMV, cytomegalovirus.

with a fluoroquinolone routinely, potentially causing differences in baseline microbiota.

In the present study, no statistical difference was seen between the two groups in regard to PFS and OS. However, there was a trend toward increased OS in the patient group who did not receive antibiotics with anaerobic coverage. It is hard to determine the exact cause of this trend as PFS and OS are multi-factorial. However, patients who did not receive anaerobic coverage were also less likely to receive an allo-HCT from an unrelated donor and develop aGVHD, both of which are risk factors for decreased survival.

There was a statistically significant difference found in TRM between the groups, with an increased TRM at 100 days and 1-year post-transplantation in patients who received antibiotics with anaerobic coverage. It is difficult to identify what specifically led to this difference. But, as mentioned above, those who received antibiotics with anaerobic coverage were more likely to receive an allo-HCT from an unrelated donor and had a greater occurrence of aGVHD, both of which can increase mortality. Peled and colleagues studied the impact of microbiota disruption and clinical outcomes after allo-HSCT. The authors obtained fecal sample data and found that higher intestinal microbiota diversity was associated with a lower risk of death. Subgroup analyses further showed an association between lower microbiota diversity and TRM and GVHD-related mortality.²² Shono and colleagues found that the use of broad-spectrum antibiotics with anaerobic coverage was associated with GVHD-related mortality while the use of aztreonam and cefepime was not in patients after allo-HSCT.¹⁴ These data parallel the findings reported in our study and provide additional reason to consider antibiotic spectrum when using antibiotics in patients undergoing an allo-HSCT.

Although this was a retrospective study that comes with inherent limitations, one unique strength of our study is that antibiotic choice was cycled and standardized, limiting treatment bias. There are, however, limitations to consider. It is non-randomized, and its single center design limits generalizability. Beyond the use of prophylactic antibiotics, we did not capture antibiotics administered before transplantation, which could have influenced microbiota diversity in these patients. Additionally, although the majority of patients in the anaerobic group only received antibiotics with anti-anaerobic coverage, some patients also received non-anti-anaerobic antibiotics targeting gram-negative organisms within the first 30 days post-allo-HSCT. We did not capture durations of therapy nor were we able to assess this cross-over effect on the development of aGVHD. This may make

FIGURE 2 Progression-free survival based on antibiotic use (anaerobic vs non-anaerobic coverage)

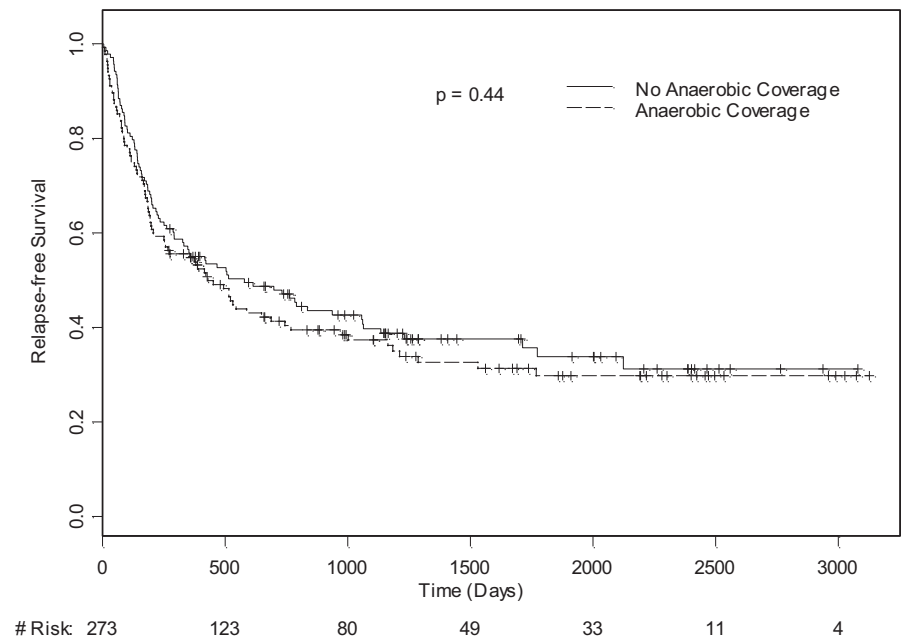
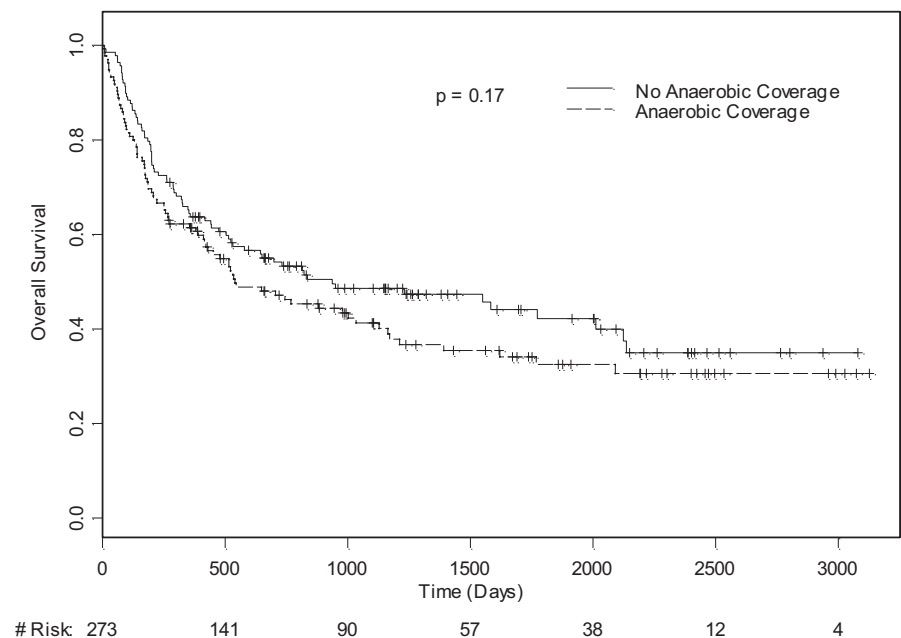


FIGURE 3 Overall survival based on antibiotic use (anaerobic vs non-anaerobic coverage)



it difficult to conclude a strong causal relationship between receiving anti-anaerobic coverage and development of aGVHD. Lastly, we did not capture rate of chronic GVHD, but mechanisms by which it occurs differ from aGVHD and would likely not be as affected by early antibiotic use.

5 | CONCLUSION

Patients who received antibiotics with anaerobic coverage during the first 30 days after an allogeneic HSCT appeared to be at an increased risk of developing aGVHD and had a higher rate of TRM. This observation is concerning that anaerobic coverage may be

contributing to aGVHD presumably by fostering drug-associated gut dysbiosis. Of course, other factors like CMV reactivation and type of allogeneic HSCT may also contribute to the risk of developing aGVHD. The current study provides some initial real-world data on the relationship between antibiotic use and its association with aGVHD in patients undergoing allogeneic stem cell transplantation. Larger well-controlled trials are warranted to further clarify these relationships.

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CONFLICTS OF INTEREST

No relevant conflicts of interest to disclose by any authors.

AUTHORS' CONTRIBUTIONS

CRH took part in concept design, data collection, data analysis, data interpretation, drafting manuscript, critical revisions, and manuscript approval. DS was involved in concept design, data analysis, data interpretation, critical revisions, and manuscript approval. ASK took part in concept design, critical revisions, and manuscript approval. SW contributed in concept design, statistical analysis, data analysis, critical revisions, and manuscript approval. AC performed concept design, data analysis, data interpretation, critical revisions, and manuscript approval.

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